



INTE 202 INTEGRATIVE PROGRAMMING AND TECHNOLOGIES FINAL PROJECT REQUIREMENT

Project Title:

Real-Life Data Analytics System using Python

Project Overview:

Students will work in groups to design and develop a **data analytics project** that solves or analyzes a **real-life problem, process, or system** using Python. The project must demonstrate the application of the fundamental concepts of data analytics, including:

- Data collection and preparation
- Data cleaning and preprocessing
- Data manipulation
- Statistical and numerical computation
- Data visualization
- Insight generation and interpretation

The project should use a **real dataset** or a realistically simulated dataset related to an actual problem or scenario from industries, organizations, schools, businesses, healthcare, transportation, social media, finance, environment, or other real-world contexts.

The project must showcase how data analytics can help in decision-making, prediction, monitoring, evaluation, or improvement of a system or process.

General Requirements:

Group Composition

- Each group must consist of **7-10 members**.

Project Objectives:

The project must:

1. Apply Python programming in data analytics.
2. Analyze a real-life problem using datasets.
3. Demonstrate data cleaning and preprocessing techniques.
4. Perform meaningful computations and statistical analysis.
5. Generate graphical visualizations and insights.
6. Present conclusions and recommendations based on analyzed data.



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Required Python Libraries:

Students are encouraged to use the following libraries:

- pandas
- numpy
- matplotlib
- seaborn (optional)
- plotly (optional)
- scikit-learn (optional for predictive analytics)

Suggested Project Topics:

Groups will identify and propose their own topic. The topic must involve a real-life problem or system that can benefit from data analysis. Examples include:

- Student academic performance analytics
- Sales and inventory analysis
- Hospital patient monitoring analytics
- Social media sentiment analysis
- Public transportation delay analysis
- Food delivery performance analytics
- Online shopping customer behavior analysis
- Weather trend analysis
- Employee attendance and productivity analytics
- Fitness and health tracking analytics
- Traffic accident analysis
- Financial expense monitoring system
- Water or electricity consumption analysis
- Tourism trend analysis
- Pollution monitoring analytics

Note: Groups may also propose other approved topics.

Dataset Requirements:

The dataset must:

- Contain at least **500 rows** of data
- Include at least **6 attributes/columns**
- Contain numerical and/or categorical data
- Include missing values, inconsistent entries, duplicates, or noisy data that require cleaning
- Be stored in formats such as:
 - .csv
 - .xlsx
 - .json

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Datasets may come from:

- Kaggle
- Government open-data portals
- School-generated surveys
- Business records
- Public APIs
- Simulated realistic datasets

Required Project Components:

1. Problem Definition

The group must:

- Explain the real-life problem or system
- Identify stakeholders/users
- Explain why data analytics is useful for the problem

2. Dataset Description

Discuss:

- Source of the dataset
- Meaning of each column/attribute
- Number of records
- Data types
- Possible issues in the dataset

3. Data Cleaning and Preprocessing

The project must demonstrate data cleaning techniques such as:

- Handling missing values
- Removing duplicates
- Correcting inconsistent entries
- Renaming columns
- Converting data types
- Filtering invalid data
- Handling outliers
- Normalization or standardization (optional)

Students must explain:

- What problems were found
- What cleaning methods were applied
- Why the methods were necessary



4. Data Manipulation and Processing

The project must include operations such as:

- Sorting
- Filtering
- Grouping
- Aggregation
- Merging datasets (if applicable)
- Calculating totals, averages, percentages, rankings, trends, or correlations

Examples:

- Monthly sales totals
- Average student grades
- Customer spending behavior
- Most common diseases
- Traffic congestion trends

5. Statistical and Analytical Computations

The project must include meaningful computations such as:

- Mean
- Median
- Mode
- Standard deviation
- Minimum and maximum values
- Correlation analysis
- Trend analysis
- Frequency distributions

Optional advanced analytics:

- Predictive analysis
- Forecasting
- Classification
- Clustering

6. Data Visualization

The project must include at least:

- 1 Bar Graph
- 1 Line Graph
- 1 Pie Chart
- 1 Histogram or Scatter Plot



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Visualizations must:

- Be properly labeled
- Include titles
- Use readable formatting
- Help explain insights from the data

7. Insights and Recommendations

Students must:

- Interpret the results
- Identify trends or patterns
- Explain findings
- Provide recommendations or solutions based on the analysis

Example:

- “Sales are highest during weekends.”
- “Students with low attendance tend to have lower grades.”
- “Traffic congestion peaks during rush hours.”

System/Application Requirement:

The group must create a Python-based analytics application that can:

- Load datasets
- Display cleaned data
- Perform computations
- Generate charts automatically
- Output summarized results

The application may be:

- Console-based
- GUI-based using Tkinter
- Web-based (optional for advanced groups)

Required Outputs:

Each group must submit:

A. Source Code

- Complete Python files (.py)
- Properly commented code

B. Dataset

- Original dataset
- Cleaned dataset

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C. Documentation/Final Paper

Must include:

- | | |
|--------------------------|-----------------------------------|
| 1. Title Page | 8. Data Analysis and Computations |
| 2. Introduction | 9. Visualizations |
| 3. Problem Statement | 10. Findings and Insights |
| 4. Objectives | 11. Conclusion |
| 5. Dataset Description | 12. Recommendations |
| 6. Methodology | 13. References |
| 7. Data Cleaning Process | |

D. Presentation

Groups must present:

- The problem
- Dataset
- Cleaning methods
- Analysis performed
- Visualizations
- Insights and recommendations
- Live demonstration of the program

Suggested Development Flow:

- | | |
|------------------------------|----------------------------|
| 1. Topic Proposal | 6. Visualization |
| 2. Dataset Collection | 7. Application Development |
| 3. Data Cleaning | 8. Documentation |
| 4. Exploratory Data Analysis | 9. Final Presentation |
| 5. Statistical Computation | |

Grading Rubric:

Criteria	Percentage
Relevance and Complexity of Topic	10%
Dataset Quality and Appropriateness	10%
Data Cleaning and Preprocessing	20%
Data Manipulation and Computation	20%
Data Visualization	15%
Insights and Recommendations	10%
Program Functionality and User Interface	10%
Presentation and Documentation	5%
Total	100%



Additional Notes:

- Groups are encouraged to use creative and meaningful datasets.
- Plagiarism is strictly prohibited.
- Projects must demonstrate originality and proper application of data analytics concepts.
- Groups should ensure that interpretations and conclusions are supported by data.
- Proper coding practices and documentation are expected.

Expected Learning Outcomes:

By the end of the project, students should be able to:

- Apply Python in solving real-world data problems
- Clean and preprocess datasets effectively
- Analyze and interpret data
- Create meaningful visualizations
- Develop simple analytics systems
- Present data-driven insights professionally

Prepared by:

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